CMAN :=

Given data, N=32, FP32 = 4 bytes, DRAM bandwidth = 320GB/s
compute = 10TFLOPS.

Task 1 :=

N=32 times, so each B[K][J] accessed 32 times

A access $N^3 = 32^3 = 32,768$

B access $N^3 = 32^3 = 32,768$

Total elements access $= 2N^3 = 2 \times (32)^3 = 65,536$

Total DRAM units $= 2N^3 \times 4 = 65,536 \times 4$

$= 262,144$ bytes $\simeq 256KB$

$\boxed{\text{Total DRAM traffic} = 256KB}$

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Task 2 :-

Tiled size $T=8$

$\left(\frac{N}{T}\right)^2 = \left(\frac{32}{8}\right)^2 = 16$ tiles per matrix

each tile holds $(T^2) = (8)^2 = 64$ elements

$= 64 \times 4$ bytes

$= 256$ bytes

Dram loads for $A = N^2 \times 4$

$= (32)^2 \times 4 = 4096$ bytes

total Dram for A and B

$= N^2 \times 4 \times 2$

$= 4096 \times 2 = 8192$ bytes

Dram traffic = total Dram load for A and B is 8192 bytes.

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Task 3 :=

Traffic ratio :=

Naive traffic = 262,144 bytes
Tiled traffic = 32,768 bytes

$$\text{Traffic ratio} = \frac{\text{Naive}}{\text{Tiled}} = \frac{262,144}{32,768} = 8 \ (T)$$

Tiled traffic = Naive traffic ÷ N

= 262,144 ÷ 32

Tiled traffic = 8,192 bytes

$$\text{Ratio} = \frac{262,144}{8,192} = 32 = N$$

- In the naive traffic, we reload matrix B from memory 32 times because every row of A needs to read all of B again. Tiling fixes this by keeping B in fast memory and reusing it, so we only load B once. That is why naive method uses 32 times more memory traffic exactly N times more.

Task 4 :=

Bandwidth = 320 GB/s
compute = 10 TFLOPS
Total FLOPs = $2 \times N^3 = 2 \times 32^3 = 65,536$ FLOPs.

Naive case :=

t_memory = Traffic / BW

= 262144 / (320 GB/s)

= 8.192×10^{-7} s = 0.819 μs

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Task 4

Naive case:

Number of FLOPs $= 2 \times N^3 = 65,536$ FLOPS

Total Dram traffic $= 2 \times N^3 \times 4$
$= 262,144$ bytes

$$AI = \frac{65,536}{262,144} = 0.25 \text{ FLOP/byte}$$

Tiled case:

Number of FLOPs needed $= 2N^3 = 65,536$ FLOPs

Total Dram traffic $= 2 \times N^2 \times 4 = 8192$ bytes

$$AI = \frac{65,536}{8192} = 8 \text{ FLOPs/byte.}$$

Ridge point:

B.W $= 320$ GB/S, peak compute $= 10$ TFLOPS

$$rp = \frac{\text{peak compute}}{BW} = \frac{10 \times 10^{12}}{320 \times 10^{9}} = 31.25 \text{ FLOP/byte}$$

for both naive case and tiled case the bottleneck is memory bound.

Execution time for Naive case:

$$\text{time for computation} = \frac{\text{FLOPS needed}}{\text{Peak Compute}}$$

$= 655$ ns

$$\text{time for memory} = \frac{\text{memory}}{\text{peak BW}} = 8192 \times 10^{-10} \text{ secs}$$

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Execution time for Tiled case

Time for computation = 6.55 ns.

time for memory $= \frac{8192}{320 \times 10^9} = 25.6$ ns

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